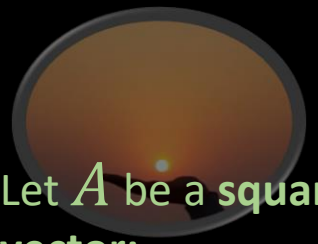
Eigen Vector
And
Eigen value



Eigen vector and Eigen value

Let A be a **square matrix** of size $n \times n$ and let **v be a non-zero vector:**

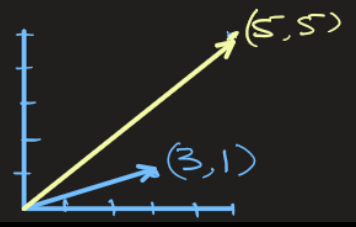
The operation A**v** transform v → Av.

A is also called transformation matrix.

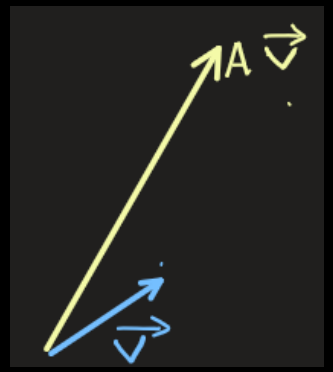
Imagine a matrix A as something that **transforms space** — it can **rotate, stretch, flip, or squash** vectors.

Let $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$, $\vec{v} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

So, $A\vec{v} = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1\times3 + 2\times1 \\ 2\times3 - 1\times1 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$

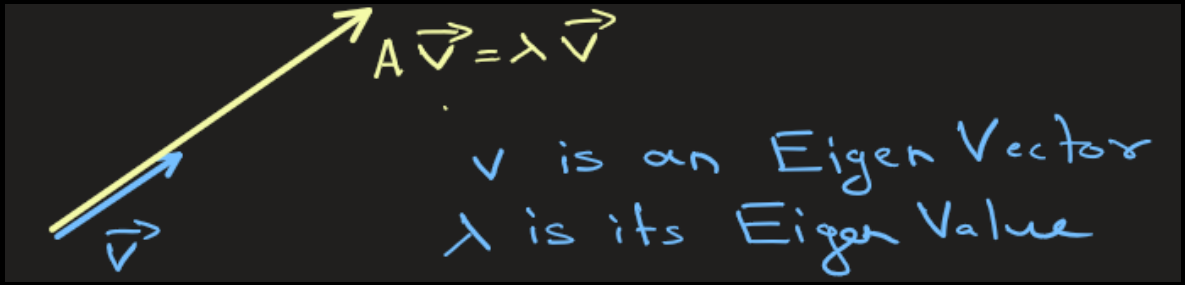


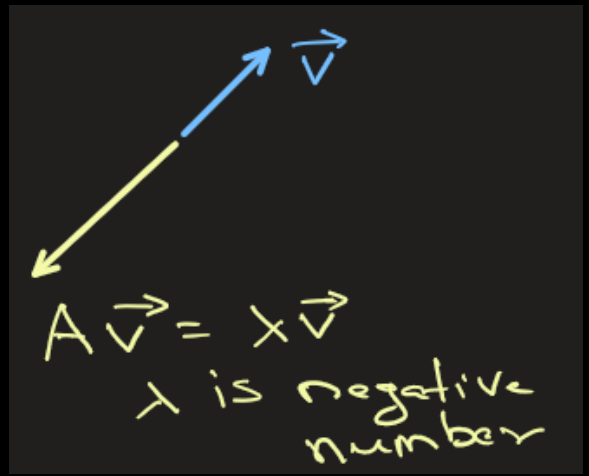
So, when a matrix A acts on a vector, it generally **transforms** it — the resulting vector usually **changes direction(i.e. gets knocked off its axis)** *and* **changes magnitude** compared to the original.



Eigen vector and Eigen value

Some special vectors do **not change their direction** when a matrix transforms them.
These are called **eigenvectors**.
Although their **direction stays the same**, their **length** (magnitude) may change — they can be **stretched** or **compressed**.
The **amount of stretching or shrinking** is given by a number called the **eigenvalue (λ)**





Side Note: The word "**direction**" above means the *line or axis through the origin.*
Eigen vectors stays on the *same line* through the origin, but may get reversed.
In that sense, reversing does not change direction — it's still along the same axis

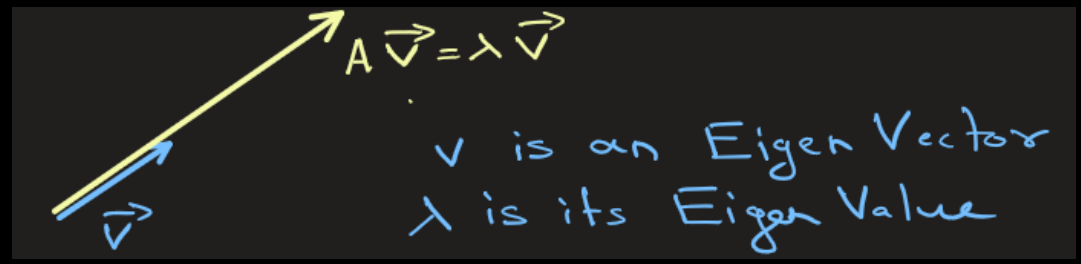
Eigen Vector And Eigen Value: Definition

Let A be a **square matrix** of size $n \times n$.

A **non-zero vector v** is called an **eigenvector** of A if it satisfies the equation:

A **v** = λ **v**

where λ is a **scalar** (a single number), called the **eigenvalue** corresponding to **v.**

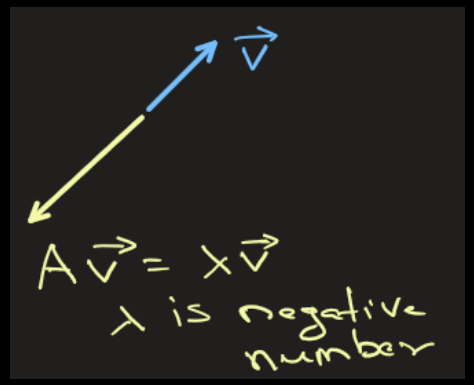


Key points:

A: transformation matrix

v : direction that **does not change** under A **(It only stretched or shrunk)**

λ: factor by which **v** is stretched (or flipped if negative)



Example intuition:

Think of A as a rubber sheet transformation.

Most points move in weird direction: i.e . they change their axis.

But some points move along **certain straight lines** - not sideways, i.e they move **back and forth.**

These lines correspond to **eigenvectors**,

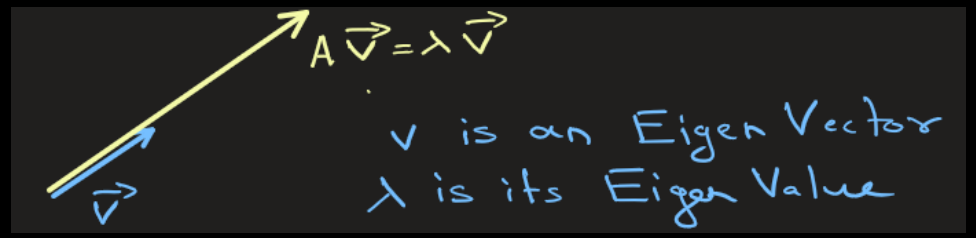
and how far they move (multiplied by λ) tells you the **eigenvalue**.

How do You find Eigen Vectors and Eigen Values

1) Form the characteristic matrix:

$$A - \lambda I,$$

where I is the identity and λ is a scalar.



2) Characteristic polynomial: Compute the determinant det(A−λI).

3) Find eigenvalues: Solve

det(A−λI) = 0 for λ.

These roots are the eigenvalues.

4) Find eigenvectors: For each eigenvalue λ_i, solve the linear system $(A - \lambda_i I)\mathbf{v} = \mathbf{0}$ for non-zero vectors $\mathbf{v}$. Any nonzero solution is an eigenvector for λ_i.

5) (Optional) Normalize eigenvectors if you want orthonormal basis vectors. If A is symmetric, eigenvectors corresponding to distinct eigenvalues are orthogonal.

6) (Optional) Diagonalize:
If you collect eigenvectors into a matrix P(columns are eigenvectors) and D is diagonal of eigenvalues, then $A = PDP^{-1}$(and if eigenvectors are orthonormal, $P^{-1} = P^T$)

How do You find Eigen Vectors and Eigen Values: Example

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Step 1 — **Form the characteristic matrix:** $A - \lambda I$

$$A - \lambda I = \begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix}$$

Step 2 — characteristic polynomial (determinant):

Compute det(A−λI):
det(A − λI)
= (2 − λ)(2 − λ) − 1
= λ^2− 4λ + 3

Step 3 — solve $\lambda^2 - 4\lambda + 3 = 0$:
$\lambda^2 - 4\lambda + 3$ = 0
$\Rightarrow$ (λ − 1)(λ − 3) = 0
$\Rightarrow$ λ_1= 1, λ_2=3

How do You find Eigen Vectors and Eigen Values: Example

Step 4 — find eigenvectors

- For $\lambda_1 = 3$, solve $(A - \lambda_1 I)\mathrm{v} = \mathbf{0}$:

$$(A - \lambda_1 I)\vec{v}_1 = 0$$

$$\left(\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - 3\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right)\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \left(\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}\right)\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2-3 & 1-0 \\ 1-0 & 2-3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -x_1 + x_2 \\ x_1 - x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \quad x_1 = x_2$$

pick $x_2 = 1$.

so $\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

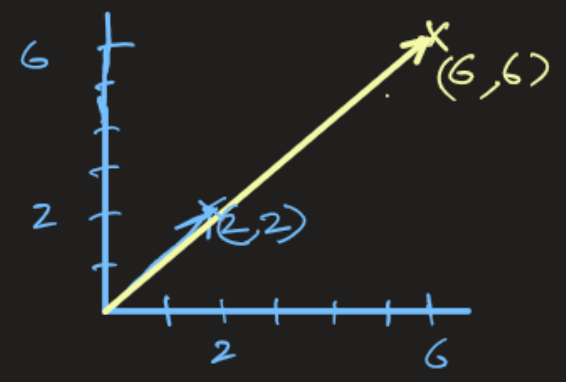
Here all vectors that span along (1,1) gets scaled by Matrix A by factor of 3. These scaled vectors do not change direction.

For example, if **v** = (2,2) → A**v** = 3**v** (see fig.)

$$\vec{v} = (2,2) = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$$A\vec{v} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 6 \end{bmatrix} = 3\begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$$= 3\vec{v}$$



How do You find Eigen Vectors and Eigen Values: Example

- For $\lambda_2 = 1$,solve $(A - 1I)\mathbf{v} = \mathbf{0}$:

$$(A - \lambda_2 I)\vec{v}_2 = 0$$

$$\left(\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - 1 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right)\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\left(\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right)\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2-1 & 1-0 \\ 1-0 & 2-1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_1 + x_2 \\ x_1 + x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \quad x_1 = -x_2$$

pick $x_2 = -1$

so $\vec{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Here all vectors span along (1,-1) gets scaled by Matrix A by factor of 1. These scaled vectors do not change direction.

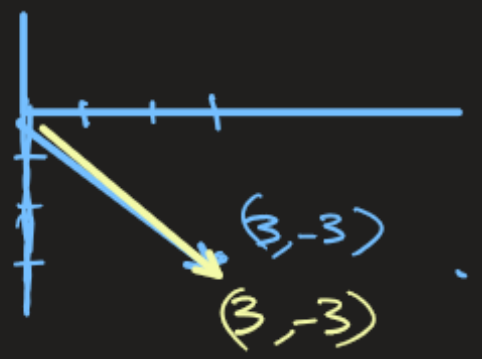
For example, if **v** = (3,-3) → A**v** = 1**v** (see fig.)

$$\vec{v} = (3, -3) = \begin{bmatrix} 3 \\ -3 \end{bmatrix}$$

$$A\vec{v} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 3 \\ -3 \end{bmatrix} = \begin{bmatrix} 3 \\ -3 \end{bmatrix}$$

$$= 1 \cdot \begin{bmatrix} 3 \\ -3 \end{bmatrix}$$

$$= 1\vec{v}$$



How do You find Eigen Vectors and Eigen Values: Example

Verify: Av = λv

$$Av_1 = \lambda_1 v_1$$

$$Av_2 = \lambda_2 v_2$$

- $\mathbf{A}\mathbf{v}_1 = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$ ✓
- $\mathbf{A}\mathbf{v}_2 = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$ ✓

How do You find Eigen Vectors and Eigen Values: Example

(Optional) Normalize Eigen Vectors:

Normalize v_1:

$$\hat{\vec{v}}_1 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Normalize v_2:

$$\hat{\vec{v}}_2 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

(Optional) — diagonalization (since A is symmetric)
Take orthonormal eigenvectors as columns of P:

$$P = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}, \qquad D = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}.$$

Because P is orthogonal (i.e. $P^{-1} = P^T$) , we have $A = PDP^T$.
You can verify this last equation:

$$A = PDP^T$$

$$= \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = \underline{\underline{A}}$$

EXTRA

Key Properties of Eigen Vectors

Eigen Vectors are not generally **orthogonal**, but they are orthogonal for a particular type of matrix such as a symmetric matrix.

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \Rightarrow$$

symmetric matrix

Eigen Vectors

$\vec{v}_1 = (1, 1)$

$\vec{v}_2 = (1, -1)$

$\vec{v}_1 \cdot \vec{v}_2 = 1 \times 1 - 1 \times 1$

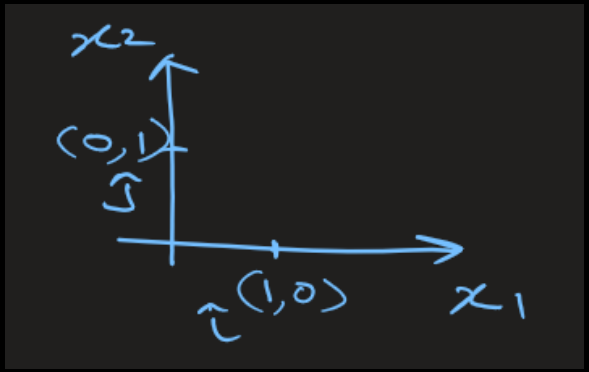
$= 0$

(v_1 & v_2 are orthogonal)

Let's see the effect of A on the **basis** vectors:

Let $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$, $\vec{v} = \hat{i} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

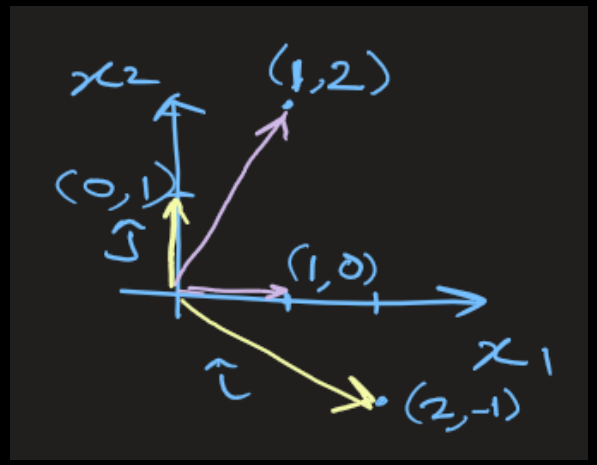
↳ basis vector along x_1-axis



$$A\vec{v} = A\hat{i}$$
$$= \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$
$$= \begin{bmatrix} 1 \\ 2 \end{bmatrix} \equiv A_1 \quad (\text{So, } (1,0) \xrightarrow{A} (1,2))$$

Similarly if $\vec{v} = \hat{j} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$A\vec{v} = A\hat{j}$$
$$= \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$
$$= \begin{bmatrix} 2 \\ -1 \end{bmatrix} \equiv A_2 \quad (\text{So } (0,1) \xrightarrow{A} (2,-1))$$



So I can rewrite vector A as

$$A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} = [\underbrace{A_1}_{\text{transforms } \hat{i}} \quad \underbrace{A_2}_{\text{transforms } \hat{j}}]$$

A_1 A_2

Conclusion: The **columns of a matrix A** tell you **how the basis vectors transform** under that matrix.

Example of 3X3 matrix: (p1)

🎯 Let's pick this matrix

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 4 \\ 0 & 4 & 9 \end{bmatrix}$$

👉 I picked this because its eigenvalues and eigenvectors are easier to compute but still show full 3×3 process.

📌 Step 1: Characteristic equation

$$\det(A - \lambda I) = 0$$

$$A - \lambda I = \begin{bmatrix} 2-\lambda & 0 & 0 \\ 0 & 3-\lambda & 4 \\ 0 & 4 & 9-\lambda \end{bmatrix}$$

📌 Step 2: Compute determinant

Since row 1 has only one nonzero entry:

$$= (2-\lambda) \cdot \det \begin{bmatrix} 3-\lambda & 4 \\ 4 & 9-\lambda \end{bmatrix}$$

Compute 2×2 determinant

$$= (2-\lambda)\left((3-\lambda)(9-\lambda) - 4 \cdot 4\right)$$

$$= (2-\lambda)\left((3-\lambda)(9-\lambda) - 16\right)$$

Expand (3 - λ)(9 - λ)

$$= (3-\lambda)(9-\lambda) = 27 - 3\lambda - 9\lambda + \lambda^2 = \lambda^2 - 12\lambda + 27$$

So:

$$= (2-\lambda)\left(\lambda^2 - 12\lambda + 27 - 16\right) = (2-\lambda)(\lambda^2 - 12\lambda + 11)$$

Example of 3X3 matrix: (p2)

📌 Step 3: Solve characteristic equation

$$(2-\lambda)(\lambda^2 - 12\lambda + 11) = 0$$

✅ First eigenvalue:

$$\lambda_1 = 2$$

✅ Solve quadratic:

$$\lambda^2 - 12\lambda + 11 = 0$$

Use quadratic formula:

$$\lambda = \frac{12 \pm \sqrt{12^2 - 4 \cdot 1 \cdot 11}}{2} = \frac{12 \pm \sqrt{144 - 44}}{2} = \frac{12 \pm \sqrt{100}}{2} = \frac{12 \pm 10}{2}$$

$$\lambda_2 = \frac{12+10}{2} = 11 \quad \lambda_3 = \frac{12-10}{2} = 1$$

✅ Final eigenvalues

$$\boxed{1,\ 2,\ 11}$$

📌 Step 4: Find eigenvectors

For λ = 2

$$(A - 2I) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & 4 & 7 \end{bmatrix}$$

👉 First row → automatically satisfied (0=0)

From other rows:

$$1y + 4z = 0 \Rightarrow y = -4z$$

$$4y + 7z = 0$$

Substitute y = -4z:

$$4(-4z) + 7z = -16z + 7z = -9z = 0 \Rightarrow z = 0 \Rightarrow y = 0$$

✅ Eigenvector:

$$\vec{v} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Example of 3X3 matrix: (p3)

For λ = 11

$$(A - 11I) = \begin{bmatrix} -9 & 0 & 0 \\ 0 & -8 & 4 \\ 0 & 4 & -2 \end{bmatrix}$$

1st row → x = 0

From 2nd:

$$-8y + 4z = 0 \Rightarrow 2z = 4y \Rightarrow z = 2y$$

✅ Eigenvector:

$$\vec{v} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

For λ = 1

$$(A - I) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 4 \\ 0 & 4 & 8 \end{bmatrix}$$

1st row → x = 0

From 2nd:

$$2y + 4z = 0 \Rightarrow y = -2z$$

✅ Eigenvector:

$$\vec{v} = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

✅ Final results

Eigenvalue	Eigenvector
2	$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$
11	$\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$
1	$\begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$

✨ How to explain in class

👉 Each eigenvector shows a direction the matrix stretches (or shrinks) by its eigenvalue.

👉 Eigenvalue tells how much stretching (or compressing).

✅ Example:

- Along (1,0,0) the matrix scales by 2
- Along (0,1,2) scales by 11
- Along (0,-2,1) scales by 1